

Jacob Currie

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Data Analysis and Visualizations

Data Exploration and Analysis | Workflow Automation | Visualization

Experienced detail-oriented professional with a strong academic background in Data Engineering, Data Analysis and Geographic Information Systems. Highly motivated to uncover the hidden value in data with Machine Learning and Business Intelligence tools for clear decision making, accurate forecasting, and deeper more powerful insights.

Skilled in spatial and non-spatial data engineering and analysis techniques using Python, SQL, Power BI, Excel, and ArcGIS Pro for data driven decisions and solutions. Experience constructing models and scalable ETL/ELT pipelines to gather data from diverse sources in various formats, explore data and perform transformations, and load into target tables for Machine Learning, further analysis, or output as finished products. Experience creating clear insightful visualizations for analysis and Business Intelligence reporting. Familiar with designs of OLAP databases for Data Warehouses and OLTP databases for transactional operational data.

Demonstrated ability to deliver accurate, high-quality results in academic and professional settings. Enjoys collaborating with others to provide data related solutions and leveraging technology for effective efficient analysis. Committed to life-long learning of data technologies and sharing knowledge with others to strengthen abilities throughout the organization.

EDUCATION & CERTIFICATIONS

Austin Community College - *Austin, Texas*

Geographic Information Systems - Level 2 Certificate (2-Year)

Texas State University - *San Marcos, Texas*

Bachelor of Science - Geography, Water Resources

Udemy Courses Completed:

SQL and PostgreSQL: The Complete Developer's Guide – Stephen Grider

The Complete SQL Bootcamp – Jose Portilla

Python Data Science: Data Prep & EDA with Python – Maven Analytics

Microsoft Power BI Desktop for Business Intelligence – Maven Analytics

WORK EXPERIENCE

Austin Community College - *Austin, Texas*

April 2023 – May 2023

Intern

Worked with a team of interns to complete a water infrastructure project contracted by the City of Round Rock. I used imagery and landscape reference points to locate and digitize water meters and associated water lines with ArcGIS Pro.

- Successfully located and digitized over 500 water meters to help complete the project on time.

Sky Ranch Terlingua - *Terlingua, Texas*

June 2018 - Present

Owner and Operator

Designed and built multiple campsites and structures in the Chihuahuan Desert north of Big Bend National Park. Manage ongoing operations.

- Located suitable land parcels based on accessibility and area views.
- Designed and built structures to withstand a high wind desert environment.
- Planned construction methods so that all structures could be built by one person.
- Generates an average annual return on investment of 27%.
- Successfully tracked bookings in relation to promotional pricing timing to determine how far ahead to send discounted price notifications which led to over 10% increase in bookings.

Data Foundry, Inc. – *Austin, Texas*

June 2009 – August 2018

Assistant Facility Manager

Facility Department Coordinator

Data Center Technician

Coordinated data center infrastructure maintenance events and planned technician work in a 'zero downtime' environment.

- Planned daily technician work schedules to decrease equipment downtime and increase reliability. This also allowed for forecasting labor demands.
- Provided reporting on facility operation Key Performance Indicators.
- Quantified annual labor and equipment maintenance times and projections.
- Developed facility training program and trained new technicians, reducing site restoration time from loss of power and equipment failure events by over 40%.
- Worked with vendor to roll out a new Computerized Maintenance Management System which centralized equipment documentation, logs, tickets, and repair times.
- Developed comprehensive preventive maintenance program for Austin sites to ensure all equipment was properly maintained within scheduled windows.
- Managed documentation project and wrote procedural documents for Emergency Operation and Maintenance Operation of electrical and mechanical systems, equipment trains, and individual equipment.

PROFESSIONAL SKILLS

- ETL/ELT pipelines with Python and SQL
- Data Exploration and Analysis with Excel, SQL, Python, Pandas, and GeoPandas
- Creating visualizations with Python and Business Intelligence tools such as Power BI
- Creating reports and dashboards
- ESRI Products: ArcGIS Pro and ArcGIS Online
- Geospatial Analysis such as Suitability/Site Analysis, Overlay, Proximity
- Cartography, including choosing appropriate map projections, visual hierarchy
- Digitizing and Editing Features
- Model Builder and Custom Geoprocessing Tool Creation
- Experienced with Python, Arcade, SQL, Data Interoperability Extension (FME)
- Familiar with writing Python scripts using Pandas and GeoPandas DataFrames along with PostgreSQL (OLTP) and DuckDB (OLAP) databases for data manipulation and efficient queries
- Critical Thinking and Problem Solving
- Communication with others from varied backgrounds and training
- Defining Stakeholder Requirements
- Exploring Use Cases
- Knowledge of Geography, Hydrology, Fluvial Processes, Water Resources

INTERESTS

- Finding new ways to use emerging technology to deliver more data products in less time with greater detail
- Providing high-value data services and products to stakeholders
- Streamlining data analysis work with customized automated ETL tools
- Expanding knowledge of SQL, Python, Workflow Automation and Reporting
- Data Science and Engineering
- Using AI, ML, DL models and methods to extract actionable intelligence
- Big Data Analysis
- Efficient Automation of Analytical Processes
- Automation of Analysis to provide updated datasets
- Building Custom Map Applications and Analysis Tools
- Quantifying phenomena for analysis
- Using Cloud Compute Resources for Analysis and Creating Datasets